THE COMPARISON BETWEEN THE COMPLETION ALONG A CLOSED SUBSPACE AND ALONG ITS UNDERLYING CLOSED SUBSET IS AN ISOMORPHISM ONLY UNDER A FINITE PRESENTATION HYPOTHESIS

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains no diagram; its content is prose and inline formulas, reproduced verbatim, including the counterexample with the polynomial ring in countably many variables.

Editorial note. The counted spans carry 3 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Both completion constructions commute with etale localisation. Same chapter, the locality of the two completions, used at proof lines 8317-8320. Reduces the assertion to the affine case.
- The completion of an affine scheme along a closed subset, and along a closed subspace. Same chapter, Lemma lemma-affine-formal-completion-types at proof lines 8322-8325 and Lemma lemma-affine-formal-completion-subspace-types at proof lines 8322-8325. Identify both completions with the formal spectrum of the adic completion in the finitely generated case.

Source reference. Original lemma 0GXX: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 8271–8271. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GBA: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 3276–3276. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let $Z \subset X$ be a closed subspace and set $T = |Z|$. The canonical morphism $c : X^\wedge_Z \to X_{/T}$ is an isomorphism if $Z \to X$ is of finite presentation, but not in general.*

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Proof. Both constructions commute with étale localization, hence it suffices to prove this when X is affine. Say $X = \mathrm{Spec}(A)$ and Z corresponds to the ideal $I \subset A$. If I is finitely generated then both $X_Z^\wedge$ and $X_{/T}$ are equal to $\mathrm{Spf}(A^\wedge)$ where $A^\wedge$ is the I-adic completion of A, see Lemmas 0GBA and 0GXX. If $A = \mathbf{Z}[x_1, x_2, \ldots]$ and $I = (x_1, x_2, \ldots)$ then $\mathrm{Spec}(A/(x_n^n, n \geq 1)) \to X$ factors through $X_{/T}$ but not through $X_Z^\wedge$ and hence c is not an isomorphism. $\square$